

**DEPARTMENT OF SOFTWARE ENGINEERING
BACHELORS IN SOFTWARE ENGINEERING**

Complex Engineering Activity

**Course Code: CT-157, Course Title: Data Structure Algorithm and Applications CLO:
CLO 2**

Course Learning Outcome

CLO 2: Compare and contrast different data structures and algorithms to select the most suitable ones for specific real world scenarios. (Taxonomy level C4, Analysis), PLO2 (Problem Analysis)

Complex problem solving attributes covered (as per PEC - OBA manual – 2019)

- Range of resources: Involve the use of diverse resources (and for this purpose, resources include people, money, equipment, materials, information and technologies).
- Level of interaction: Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering or other issues.
- Innovation: Involve creative use of engineering principles and research-based knowledge in novel

Problem Statement

Modern organizations (universities, hospitals, transport systems, etc.) manage vast amounts of data. Efficient data organization, searching, and decision-making are essential. Your task is to design algorithms (not full code) for a Smart Information Management System (SIMS) that demonstrates knowledge of data structures and algorithms, along with their time-space complexity trade-offs. Each group must select an application context (e.g., student records, patient records, transport routes, or

Requirements (Algorithm-Oriented)

Select any 3 of the following data structures in your application. You are allowed to use different data structures across multiple features

1. Arrays & Complexity Analysis

- Design an algorithm to insert, delete, and traverse elements in a linear array.
- Write searching algorithms (linear search and binary search).
- Write sorting algorithms (Bubble Sort and Quick Sort).
- Provide complexity analysis (best, average, worst cases).

2. Stack & Recursion

- Write algorithms for infix postfix conversion and postfix evaluation using stack operations. – • Write an algorithm for the Tower of Hanoi problem (recursive). –
- Analyze the number of recursive calls and time complexity.

3. Queue Algorithms

- Write algorithms for enqueue and dequeue operations in a simple queue.
- Extend to circular queue or priority queue and write corresponding algorithms.
- Analyze efficiency of each.

4. Linked Lists

- **Basic Linked List Operations**

Write algorithms for:

- **Insertion** (at beginning, at end, at a specific position) ○ **Deletion** (from beginning, from end, from a specific position) ○ **Traversal** of a linked list • **Types of Linked Lists**

Write algorithms for *any one* of the following (or more if required):

- **Doubly Linked List** operations (insert, delete, traverse)
- **Circular Linked List** operations ○ **Singly Linked**
- **Searching in Linked Lists** Write algorithms for:
 - **Linear search** in a linked list
 - Explanation of why binary search is not directly applicable without modifications
- **Complexity Analysis**
Time & space complexity of each operation

5. Trees (Hierarchical Structures)

- Write an algorithm to insert and delete nodes in a binary tree.
- Write algorithms for in-order, pre-order, and post-order traversals.

6. Graph Algorithms

- Write an algorithm to insert and delete nodes/edges in a graph (adjacency matrix or adjacency list representation).
- Write algorithms for DFS and BFS traversals.
- Analyze the time complexity of each graph algorithm.

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SE Batch 2024, Fall Semester 2025

Grading Rubric Group Members:

Student No.	Name	Roll No.
S1		
S2		
S3		
S4		
S5		

CRITERIA AND SCALES			Marks Obtained			
			S1	S2	S3	S4
Criterion 1: Justification of Selected Data Structures						
2	1	0				
Clear, logical justification for all 3 selected structures; strong relevance to the problem context	Some justification provided; relevance partially explained	No justification OR structures not relevant to the problem				
Criterion 2: Algorithm Design Quality						
2	1	0				
Pseudocode is correct, complete, and logically structured for all chosen structures	Pseudocode partially correct; missing steps or clarity	Algorithms incorrect, incomplete, or missing				
Criterion 3: Complexity Analysis						
2	1	0				
Accurate time/space complexities; includes best, average, worst cases with explanation	Some correct complexities but missing cases or weak explanation	Incorrect or no complexity analysis				
Criterion 4: Application to Dataset						
2	1	0				
Algorithms correctly applied to a meaningful dataset; operations clearly demonstrated	Dataset used but explanation incomplete or partially correct	No dataset demonstration OR incorrect application				
Criterion 5: Documentation & Presentation						
2	1	0				
Well-organized, clear, professional report; includes diagrams/flowcharts	Understandable but with some clarity or formatting issues; diagrams may be missing	Poorly organized, unclear, or missing major sections				
Total Marks:						

Teacher's Signature: _____

